

EDITORS' CHOICE**PRISMA 2020****Introduction**

A revision of the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) statement is an important event. The PRISMA 2020 statement has, like its predecessor (now re-referred to as PRISMA 2009), been published in multiple journals. This paper describes the changes that have been introduced in the new checklist [1]. However, in the June issue of JCE we also publish a detailed report describing the development process of the update. [2]

By any measure, the papers represent a substantial body of work and a major refresh of the PRISMA statement. The process is familiar, but the detail merits repetition – robustness based on a broad search of the relevant scientific literature, focussing on those PRISMA 2009 items that are neglected or problematic, followed by a broad consultation with a wide groups of interested parties. As the authors report, over 200 methodologists and editors were invited to complete a survey, and half of those asked did so. This was followed by an in-person meeting in Edinburgh 2018 and further consultations as the drafting and revision process ran through to its completion. The list of contributors is impressively international and multi-professional, although the leadership, as demonstrated by the reports' lead authors and the principal sources of funding support, shows a considerable and outstanding Australian influence. One omission, acknowledged by the authors, is the absence of consumer or citizen involvement.

PRISMA 2020 also represents a considerable changing of the guard from the personnel perspective. Some of this, sadly but inevitably, was unavoidable, and it is good to see PRISMA 2020 being dedicated to those two towering contributors to PRISMA 2009, Doug Altman and Alessandro Liberati. That said, the list of authors of the statement includes many members of a new brigade of leaders within the international evidence synthesis community.

Key changes

The changes made between PRISMA 2020 and its predecessor highlight the changing times, both in representing new content that reflects scientific advancement, but also in a focus in supporting implementation. Although PRISMA 2020 comprises, like its predecessor, 27 items, this apparent parity is somewhat misleading: 6 of the items in

the recent iteration include several sub-items (10, 13, 16, 20, 23 and 24). The changes are well described in the main statement paper, but some of the changes are particularly noteworthy. One of these is an inevitable consequence of the success of the original guideline. Since PRISMA 2009, there have been many extensions and refinements aimed at addressing specific areas of research activity, such network meta-analysis and consideration of health equity. PRISMA 2020 cannot replicate or update each of these, but there is a clear intent on the part of its authors to be more wide-ranging in its scope. Noting that PRISMA 2009 was focussed squarely on systematic reviews of interventions, PRISMA 2020 aims to ensure relevance to a broader sweep of review types and research questions, including mixed methods reviews, those that evaluate theories of causation, prevalence or prognosis or those that do not set out to include meta-analysis or even synthesis. One of the refinements referred to above – an updated version of PRISMA for Abstracts is folded into the new guidelines, which will be welcomed by those for whom the forest of reporting guidelines is becoming somewhat confusing. Other related guidelines are simply referenced, with the hope that these will be updated to sit comfortably alongside the new PRISMA. The authors note that PRISMA 2020 may be used for updated or 'living' systematic reviews but acknowledge that additional considerations may apply in these circumstances.

Other changes relate to amendments of the process and methods. These include the increasing development and use of technologies to support study selection or data collection, the assessment of risk of bias and the introduction of the concept of certainty to replace and expand on previous and limited measures of 'study quality'. There are also measures to address new data sources, for example trials registries, websites and repositories, and also some of the known challenges in existing reviews, including unacknowledged changes from the protocol or registration data, non pre-specified sub-group or sensitivity analyses, inadequately described search strategies, and so on. Those who worry that reviews will not only become more complicated to conduct but also to read, may be somewhat relieved to note the appreciation that some of the new elements required may reasonably be cited in an online supplement or open access repository.

Importantly, and it seems surprising that this was not addressed in the PRISMA 2009 checklist, the revision in-

cludes a requirement to report on the competing influences of the review authors in addition to reporting the sources of funding. The PRISMA 2020 checklist does not specifically include a requirement that the competing influences of the primary researchers whose studies are included within the reviews will be reported, which may disappoint some commentators. However, this is explicitly covered in Item 10b of the accompanying Explanation and Elaboration document. [4]

PRISMA 2020 remains focussed on reporting, although as the authors indicate, many elements of review conduct will be implied by the reporting expectations. In this, PRISMA is consistent with its predecessor, but differs from the Methodological Expectations of Cochrane Intervention Reviews (MECIR), which are more extensive and address both conduct and reporting [3]. In a similar vein, PRISMA 2020 explicitly avoids the temptation to stray into evaluating the overall methodological quality of the systematic review. The authors note that there are existing tools to undertake this. [5,6]

Implementation and policy

The science of implementation has also developed at pace since the PRISMA 2009 guidelines was developed. Despite this, the authors of PRISMA 2020 lament the paucity of high certainty evidence on interventions that increase adherence to published reporting guidelines. Undeterred, and with implementation in mind, the authors of the new reports have developed aids including a fillable checklist and flow diagram templates on the PRISMA statement website (<http://www.prisma-statement.org/>), and a web application to facilitate completion of the checklist (available at <https://prisma.shinyapps.io/checklist/>). The checklist can also be exported in Word or PDF formats.

The PRISMA 2020 authors propose actions that may be undertaken by journals and publishers: including ensuring

that instructions to review authors include a strong recommendation that they use PRISMA 2020 in the preparation of reports prior to submission, and also that some journals may mandate the submission of a completed checklist with any appropriate systematic review. As Co-Editors of JCE, we commit to exploring these and further actions to support better reporting of reviews submitted to our journal, in conjunction with our Editorial Board.

Peter Tugwell
David Tovey

References

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